Role in the AI Research & Center of Excellence Team - Test Case

Intro

Hi!

Thank you very much for your participation in our recruitment process for the role in the Research and AI Lab team!

After looking at your CV and speaking with you we strongly believe you are a good candidate, and the next step is to confirm your capabilities with this hands-on challenge.

The following test has three parts:

- The **first** part will verify your NLP-specific knowledge. Here we will put weight on:
  - How you plan, structure, and prepare your experiments
  - How you implement your experiments
- The **second** part is a paper club, included to test your ability to engage with and critically reflect on current research. Here we will put weight on:
  - How you present and discuss the central concepts of the paper
  - How you connect the paper's key findings to potential practical relevance for DSV
- The **third** part is included to investigate your reasoning and experience around research and approach. Here we will put weight on:
  - Your knowledge of structuring a research task and project
  - How you plan to approach and how you reason about technical challenges
  - Your technical choices and decisions of techniques, methods and experiments

We estimate that these three parts should consume 6-7 hours of work. Hence, we give you at least 5 working days (7 calendar days) and we set the deadline after we confirmed you received the email from us. In case of personal time constraints, you are not able to deliver your results – please let us know beforehand – we can agree on a more suitable timeframe for you.

It is important that you present your code and conclusions in a structured and easy to read manner. If you have any questions, please do not hesitate to contact us!

We appreciate that these coding parts might be a challenge to accomplish in a way that fulfills the requirements 100%. If you do not manage to find solutions for all the tasks – please do not give up and send what you have done.

Good luck and have fun!

PART 1 – Data Science expertise

Data Description – “data science part dataset.7z”:

The dataset contains the two folders: *testing_data* and *training_data*. Both folders have identical structure. Each folder contains the two subfolders: *annotations* and *images*. In the folder *images* you will find a series of .png files. In the *annotations* folder you will find a series of .json files. The .json files contain data as it would have been returned from a Optical Character Recognition (OCR) service had the images been sent to the service. Each .json file has a corresponding image in the *images* folder. The .json and image files are linked by having the same name. The .json files contain the following variables for each text found (a text contains one or more words):

- **text:** The text containing one or many words bound together
- **label:** The categorization that the OCR service has given the text
- **words:** The words or tokens contained in the text
- **id:** The identifier of the text
- **linking:** The ids of the texts that the specific text is linked to
- **box**: Traditionally, in OCR, the axes of the bounding boxed start at the top left corner and go right (X) and down (Y), so coordinates represent this:

(X1,Y1)

SOMEEXAMPLEWORD

(X2,Y2)

Notes: **For this task it is preferred (but not necessary) that you use python.**

- Data: You should have received a data file called “data science part dataset.7z”.
- You are allowed to make certain assumptions to narrow down the scope of your work. Please make your assumptions explicit to us.
- This is not about achieving the highest accuracy/performance, but for you to broadly show your data science skills in the form of problem solving by coding.

Using the data you are asked to plan and conduct one or more data science experiments.
An experiment is defined as:

1. Using the *training_data* to train or fine-tune an algorithm to predict the dataset **labels** based on the **text and other elements of the data**.
   You can use any additional variable/data point as input data. Use the **labels** as the target values
2. Evaluate the model using the *testing_data*.
   Visualize, report, and attempt to explain the findings.
3. Reflect on what could be done differently / better if you were to continue.

PART 2 – Paper Club

Choose a machine learning / AI paper related to intelligent document processing and / or agentic AI. The paper does not have to be new or something that can be commercialized, but it's important that you are comfortable explaining central concepts and discussing how key findings might be relevant for DSV.

Write 5-10 lines on why you think the chosen paper is interesting and relevant. This will be used as a basis for further discussions during the interview.

<u>PART 3 – Research Case</u>

There is no coding done for this part. Instead, you are asked to think about **one** of the following Research Cases. Below we have three cases that represent technical cases you could be faced with on a daily basis at DSV. Pick one of the three cases for this part.

<table>
<tr><td rowspan="2">Case 1</td><td colspan="2">Case title: A generic model able to handle arbitrary input/output (information extraction)</td></tr>
<tr><td>Motivation and description</td><td>We’ve received a request to develop a single ML system capable of processing various types of documents and extracting different information for each type. For example:
Invoice: Extract purchase-related fields from the invoice.
Customs Document: Extract details pertaining to customs information.
Email: Extract order-related details from the email.
Rather than creating separate ML systems for each case, we aim to build a single system that can efficiently handle multiple use cases. This system should be flexible enough to process different types of documents as input and generate different label sets as output based on the case at hand.

Please consider:
How would you approach this, and which model, architecture, or learning paradigm would you use?
How would you compose data to train and benchmark this model?
What are the limitations of your approach?</td></tr>
<tr><td rowspan="2">Case 2</td><td colspan="2">Case title: How to represent bbox as input for generative models?</td></tr>
<tr><td>Motivation and description</td><td>Currently, we are running a unimodal, text-only LLM-based generative model for document processing (information extraction). The business has requested that we provide bounding boxes (bbox) to help explain the model’s predictions to the platform users.

This leads us to the following questions:
Should we find a way to represent bounding boxes as both input and output for a uni-modal (text-only) generative LLM-based model (e.g., Mistral, LLaMA, etc.)?
Or should we migrate to a multi-modal approach (combining text and vision)? If so, how can we use a prompt-based model to handle both field/entity extraction and bounding box identification?
Regardless of either 1 or 2, how would you approach this challenge and find a solution for this request?</td></tr>
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<tr><td></td><td></td><td>Please consider:
1. What techniques and methods would you use here?
2. How could we evaluate the accuracy of the solution?
3. What are the limitations and considerations around your choices?

For this task, you can assume that OCR, images, and a stripped text version of the OCR are available.</td></tr>
<tr><td rowspan="2">Case 3</td><td colspan="2">Case title: Apply RetNet architecture to information extraction.</td></tr>
<tr><td>Motivation and description</td><td>Have a thorough look at the following paper, which describes the RetNet architecture: https://arxiv.org/abs/2307.08621

Please consider in detail how the suggested model architecture, in this paper, could benefit a NER tagging based information extraction problem on OCR scanned documents, i.e. something spanning multiple pages worth of text.

1. Please prepare to describe Figure 1 and Figure 2 in this paper.
2. What is the time and memory complexity of this Architecture/Algorithm in comparison to the Vanilla Transformer Architecture/Algorithm during training and inference?
2.1. No need to go into too many details
3. Please prepare to describe what needs to be done during training and inference when using this architecture. How does training and inference differ?
4. Bonus: Could we imagine this architecture as an Encoder+Decoder architecture (seq2seq) and how would we go about doing that?
4.1. Could we imagine a cross-retention formulation and what would that look like?</td></tr>
</table>

Since Part 3 is a Research Case, we motivate that you challenge yourself. We don't expect a perfect solution, or even a completed solution, we would rather see you getting creative. We hope to simulate a "problem-solving session", where you will be driving the approach and process for reaching a technical direction and Research Plan.

Please describe your approach to the chosen Research Case, outlining how you would plan and execute a research project using a holistic methodology. Include the specific methods, techniques and assumptions made (technical aspects) you would utilize to investigate a potential solution.

We ask that you provide a document (e.g., PDF or similar) detailing your approach, research plan, technical evaluations, and any relevant points. During the interview, we would like to elaborate on these points with you.

Think about subjects such as:

- How would you approach this task, if this was given to you?
- What methods, techniques and strategies would you use for arriving at a solution?

- What would you do if your first, or second idea wouldn't show success?
- Research project planning and timeline
- How would you test your solution: benchmark, validation,
- What ingredients would you need to do your work: data, compute, ...